Thomas Ramesh

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Summary

Full-Stack & Software Engineer with expertise in building web platforms, 3D interactive applications, and real-time encrypted messaging systems. Co-Founder of Gohyred serving **5,000+ active users**. Experienced in React.js , Next.js, Node.js, Three.js, Redis, and Web Crypto API.

Education

Dr. A.P.J. Abdul Kalam Technical University **Greater Noida, UP**

Bachelor of Technology in Computer Science and Data Science *2023 – 2027*

Skills

- **Programming Languages:** JavaScript, TypeScript, C, C++, PHP
- **Web Development:** ReactJS, Next.JS, NodeJS, ExpressJS, MongoDB, MySQL, Tailwind CSS
- **Other Skills:** Git/GitHub, Supabase, Docker

Projects

Gohyred — Career Management Platform **gohyred.com**

– Co-founded and architected a comprehensive career management platform serving **5,000+ active users**, aggregating opportunities from multiple sources into a centralized, real-time dashboard.
– Created an AI-driven profile optimization feature that parses user-uploaded PDFs to extract skills and match candidates with relevant job opportunities.
– Delivered a robust, scalable backend architecture using Next.js API Routes and PostgreSQL, ensuring secure data management and fast query resolution.
– Developed a high-performance data pipeline using Redis and BullMQ to orchestrate web scraping tasks via Puppeteer, aggregating jobs across multiple platforms.

Vibly Chat — Encrypted Real-Time Messaging **vibly-chat.vercel.app**

– Architected a secure messaging pipeline using the native Web Crypto API, implementing ECDH (P-256) for key exchange and AES-256-GCM for message encryption.
– Achieved Optimistic UI updates (**0ms – 5ms sender latency**) and guaranteed end-to-end message delivery within **250ms – 550ms**.
– Integrated Pusher WebSockets to deliver a zero-latency chat experience with live typing indicators, instant read receipts, and real-time connection status.
– Designed a modular RESTful API using PHP 8+, utilizing a custom router and rate limiting to handle requests securely in a serverless Vercel environment.

Valenzy — 3D Chemistry Simulator & Virtual Lab **valenzy.vercel.app**

– Built 3D atomic and molecular models using React 19, Three.js, and React Three Fiber (R3F), rendering dynamic orbital animations and VSEPR geometries.
– Designed computational chemistry algorithms including an automated Chemical Equation Balancer, Gas Laws simulator, and a Virtual Reaction Workbench.
– Integrated Firebase Auth & Firestore for quiz leaderboard syncing, coupled with jsPDF for dynamic printable test generation.

Achievements & Certificates

- **Co-Founder of Gohyred** — Built and scaled platform serving **5,000+ active users**.
- **Open Source Contributor**, GirlScript Summer of Code (GSSoC 2025) *(July 2025 – October 2025).*
- **Hackathon Competitor**, Vespera '24 (Guru Gobind Singh Indraprastha University - GGSIPU, Delhi) & ASTRA 2025 (Delhi Technological University - DTU).
- **1st Position (Champion)**, District-Level Robo Sumo Competition — Demonstrated skills in robotics, engineering, problem-solving, and teamwork.
- **4th Position**, State-Level Robo Sumo Competition — Represented the district in state-level robotics.